Yuning Wu

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RESEARCH INTEREST

Design, build, and optimize AI driven robots to humanely and sensitively support and accompany people in labor-intensive tasks.
Human-robot interaction, robot learning, machine learning, robotics, computational design

EDUCATION

2019.8 - 2024.8 (Expected) **Carnegie Mellon University**, Pittsburgh, USA
Ph.D. in Computational Design
Dissertation: "A Human-Centered, Reinforcement Learning Driven Robotic Framework for Humanely Supporting Labor-Intensive Construction Workers"
Co-Advisors: Daniel Cardoso Llach (Computational Design), Jean Oh (Robotics)

2019.8 - 2021.5 **Carnegie Mellon University**, Pittsburgh, USA
M.S. in Machine Learning
Advisor: Katerina Fragkiadaki

2010.8 - 2017.7 **Tsinghua University**, Beijing, China
M.Arch. & B.Arch. in Architecture
Focused on how robots and AI agents can artistically and generatively build and interact with the physical space.

RESEARCH PROJECTS & ROBOTS

Rethinking Automation in Construction: A Reinforcement Learning Driven "Work Companion Rover" for Sensitively Supporting On-Site Construction Workers

Technical lead and main investigator, see [P1] & [P2]

Husky is an RL-driven "work companion rover" built for contextually assisting carpentry workers on-site. By bearing heavy loads and autonomously navigating complex, dynamic construction environments, Husky delivers and carries essential tools and materials, reducing physical strain on workers and allowing them to focus on skilled facets of their work, thus improving workflow fluency and workplace safety.

- Designed and architected a multi-sensor robot prototype from a chassis (scratch).
- Leveraged state-of-the-art (SoTA) advances in RL-based social robot navigation to enhance comfortable worker-robot encounters and interactions in active and complex indoor construction workspaces.
- Proposed a novel approach to contextually align a generically pre-trained RL model to context-specific worker behavioral patterns while effectively mitigating policy shift.
- Utilized SoTA LiDAR Odometry and Mapping (LOAM) methods for unstructured site mapping and robot state estimation.
- Customized an efficient multimodal, multi-sensor stack, utilizing the latest computer vision techniques to robustly detect and track worker trajectories.

A Distributed, Scalable RL-Driven Multi-Drone Framework for Flexible Task Cooperation on Complex Geometric Structures

Reinforcement learning research lead, see [P4]

Pixhawks is a fleet of drones driven by an RL-based multi–agent collision avoidance algorithm for conducting flexible building tasks such as brick-laying and spray-coating. The algorithmic framework enables a variable number of drones to adaptively and collaboratively execute tasks particularly related to complex-shaped geometric structures, showcasing a robotic framework for handling intricate real-world designs.

- Developed and deployed a scalable, decentralized, RL-based multi-agent collision avoidance algorithm to ensure efficient multi-drone operation within limited space.
- Customized a generalizable, curriculum-based Sim2Real pipeline using Unity MLAgents to bridge virtual training and real-world application.
- Conducted a proof-of-concept experiment with a physical drone prototype to validate the practicality and effectiveness of the framework.
- Analyzed the evolving role of human workers within the emerging technical system.

Learning Dense Reward for Contact-Rich Manipulation from Limited Human Demonstrations

Independent research, see [P3]

A novel dense reward —a continuous metric that reflects task progress— can be efficiently learned, rather than expert engineered, from multimodal sensory data (vision and force/torque) using a limited number of human demonstrations. The trained reward model can be used in RL for contact-rich robotic manipulation tasks such as USB insertion, door opening, and joint assembly.

- Designed a novel Temporal Variant Forward Sampling (TVFS) method, robust and cost-efficient in generating learning samples from human demonstrations within a physical simulation environment.
- Introduced a self-supervised latent representation learning architecture that efficiently leverages the temporal relationships within sequential sample pairs generated by TVFS.

Teaching AI Spatial Sense by Playing with LEGOs

Independent research

LEGO Player is an AI agent capable of creating stable stacked structures using LEGO blocks in NVIDIA Omniverse Isaac, supported by a trained physics inference engine that can predict the stability of multiple rigid bodies based on spatial relationships.

- Built a Graph Neural Network (GNN)-based physics engine to accurately infer the stability of stacked LEGO structures.
- Implemented hierarchical reinforcement learning (HRL) methods to effectively search for valid block stacking patterns, optimizing both efficiency and structural integrity.

Discovering Water and Ice on the Moon

System engineering team member

MoonRanger is an autonomous, suit-case-sized moon exploration rover designed to search for signs of ice and water on the lunar pole. It is scheduled to be launched through the NASA CLPS project.

- Participated in system engineering and high-level organization of robot components.
- Created visualization and coordination dashboard for efficient progress updates and system-wise issue diagnosis.

PAPERS

[P1] **Yuning Wu**, Jiaying Wei, Jean Oh, Daniel Cardoso Llach, "*Towards Human-Centered Construction Robotics: An RL-Driven Companion Robot For Contextually Assisting Carpentry Workers,*" under review, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). Link.

[P2] **Yuning Wu***, Emek Erdolu*, Jiaying Wei, Jean Oh, Daniel Cardoso Llach. "*Construction as Computer-Supported Cooperative Work: Designing a Robot for Assisting Carpentry Workers,*" in revision, in the ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW). Accepted with revisions.

[P3] **Yuning Wu**, Jieliang Luo, Hui Li, "*Learning Dense Rewards with Temporal Variant Self-Supervision,*" in the IEEE International Conference on Robotics and Automation (ICRA), Reinforcement Learning for Contact-Rich Manipulation Workshop, Philadelphia, 2022. Link.

[P4] Zhihao Fang, **Yuning Wu**, Ammar Hassonjee, Ardavan Bidgoli, Daniel Cardoso Llach, "*Towards an Architectural Framework for Distributed, Robotically-Assisted Construction: Using Reinforcement Learning to Support Scalable Multi-Drone Construction in Dynamic Environments,*" In Proceedings of the 40th Annual Conference of the Association for Computer-Aided Design in Architecture (ACADIA), Philadelphia, 2020. Link.

[P5] Omer Akin, **Yuning Wu**, "*State Space Paradox of Computational Research in Creativity,*" in the 19th Heron Island Conference on Computational and Cognitive Models of Creative Design (HI), 2019. Link.

WORK EXPERIENCE

2020.8 - 2023.12 **Carnegie Mellon University**, Pittsburgh, PA

Graduate Researcher and Technical Lead

Experienced the full research cycle of the "Rethinking Automation in Construction" project, including funding proposal, lab space setup, robot hardware purchase, robot architecture and prototyping, algorithm training and testing, and final evaluation on construction site among workers.

2021.5 - 2022.5 **Autodesk AI Lab & Robotics Lab**, San Francisco, CA

Research Intern and Collaborator

- Worked on Inverse Reinforcement Learning (IRL) methods for robotic manipulation.
- Autodesk AI Lab has endowed a generous grant to my lab after our work together, furthering exploration in the field of AI, robotics, and computational design.

2017.12 - 2019.8 **Metascope Technology**, Beijing, China

Co-founder and CTO

- Built a vision-based style recommendation system to help commercial architects form design drafts by capturing clients' spatial preferences.
- Customized DNN models for recognizing visual architectural styles.
- Leveraged statistical machine learning methods to customize client profiling models.

HONORS AND AWARDS

2023 & 2024 **Presidential Fellowship.** Carnegie Mellon University, TCS. Highest honors in CMU.

2022 **Manufacturing Futures Initiatives Fellowship.** Carnegie Mellon University.

2022 **Autodesk AI Merit Scholarship.** Autodesk AI Lab.

2021 **Manufacturing Pennsylvania Fellowship.** State of Pennsylvania.

2020 **GSA-Provost GuSH Grant.** Carnegie Mellon University.

2019 **LEGO Scholarship.** LEGO Foundation.

2017 **Distinctive Graduation Thesis.** Tsinghua University.

2011-2014 **Siyuan Exceptional Student Fellowship.** Tsinghua University.

RESEARCH TALKS,PRESENTATIONS, POSTERS

2023.5 **TCS Innovation Forum 2023 @ Cornell Tech**. "Towards Humanely Supporting Heavy Manual Labor: Leveraging AI/ML Advances to Develop a "Work Companion Robot" for Construction Workers", invited research talk.

2023.5 **Bosch Leadership and AI Team Tech Exchange**. "Rethinking Automation in Construction", robot demonstration and presentation.

2022.11 **CMU Manufacturing Futures Institute Research Seminar**. "Rethinking Automation in Construction: Robotically Supported Collaborative Work in Heavy Manual Construction", invited research talk.

2022.10 **CMU 48-620 Graduate Seminars: Situating Research**. "Designing AI-Driven Robots For Real-World Challenges and Complexities", invited research talk.

2022.10 **CMU Board of Trustees Research Exhibit.** "Collaboratively Assist, Rather Than Brutally Replace: A Reinforcement Learning Driven Robotic Framework For Supporting Construction Workers On-Site", robot demonstration and presentation.

2022.9 **Autodesk TechX Research Exchange.** "Learning Dense Reward For Enhancing Reinforcement Learning in Contact-Rich Manipulation Tasks", invited research talk.

2022.5 **ICRA Reinforcement Learning for Contact-Rich Manipulation Workshop.** "Learning Dense Rewards with Temporal Variant Self-Supervision", poster presentation and research talk.

2021.11 **CMU 48-770 Graduate Seminars: Inquiries into Machine Learning and Design.** "Love, Death, and Robots: How Reinforcement Learning Driven Robots Can be Trained to Handle Real-World Nuances", invited research talk.

TEACHING EXPERIENCE

2022 Spring & Fall **Collaborative Instructor,** (48-749) Special Topics in Computational Design: Rethinking Automation in Construction, Carnegie Mellon University. Led and mentored a group of 4-6 MSCD master students in developing research of various topics in AI and robotics, closely related to the ReAC project. Offered hands-on, practical guidance in the development and training of robot modules.

2021 Spring **Teaching Assistant,** (10-403) Deep Reinforcement Learning and Control, Carnegie Mellon University. Engaged in designing hands-on labs, assignments, and exams.

2020 Spring **Teaching Assistant,** (48-715) MSCD Pre-Thesis I, Carnegie Mellon University. Supported a group of 10-15 MSCD students in forming their master thesis research through literature review, writing, and discussion.

2020 Fall **Teaching Assistant,** (48-716) MSCD Pre-Thesis II: Topic Research, Carnegie Mellon University. Helped the MSCD cohort in developing their thesis proposal through discussions, presentations, and reviews.

2021 Spring **Teaching Assistant,** (48-718) MSCD Thesis Defense, Carnegie Mellon University. Supported the same MSCD cohort in preparing the final steps of their thesis defense. Consistently followed through the complete cycle of MSCD thesis research mentoring.

ACADEMIC AND VOLUNTEER SERVICES

2023 **Reviewer,** ACM Conference on Computer Supported Cooperative Work (CSCW) 2024.

2022 **Panel Moderator and Critic,** M.S. in Computational Design Thesis Defense 2022.

2021 **Panel Moderator and Critic,** M.S. in Computational Design Thesis Defense 2021.

REFERENCES

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